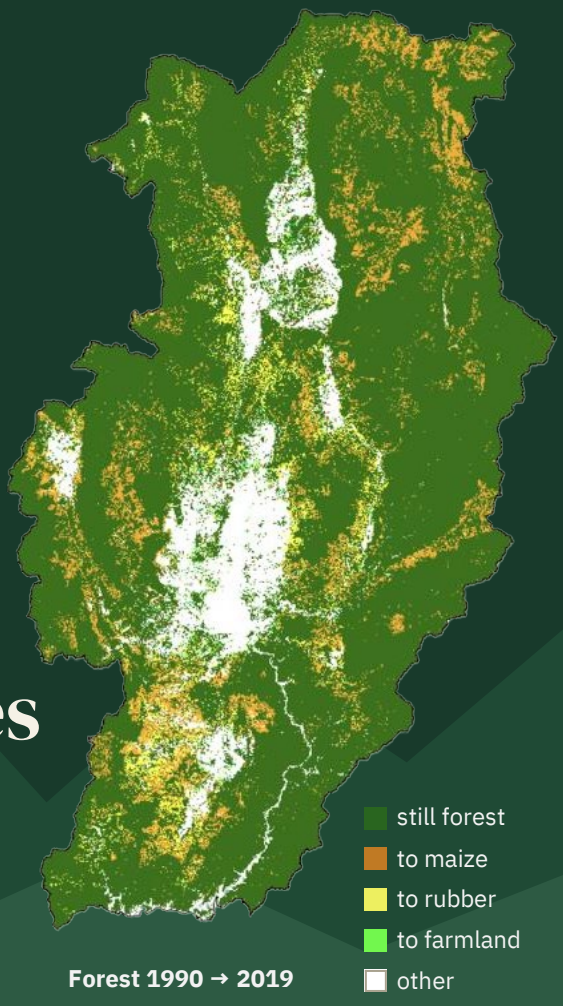
Nan

THAILAND



Forest 1990 → 2019

Kruasilp et al. 2023 [21]

Province facts and figures

Agriculture, forests, people, economy and development plans of Nan province compiled from official statistics.

11,472 km²
land area (≈7.17 million rai), 87% of it mountains

470,507
registered inhabitants, 31 Dec 2024

15 · 99 · 896
districts · sub-districts · villages

61.0%
forest cover, 2025

≈40%
of Chao Phraya runoff comes from the Nan River basin

59.5%
of employed people work in agriculture (Q3 2024)

5
ethnic groups: Tai Lue, Hmong, Yao (Mien), Htin and Mlabri ("Tong Luang").

2.6
people per household on average in 2025 Household Socio-Economic Survey [22].

8.2 yrs
average schooling of people 15+ (2024), 12th of 17 northern provinces

271
doctors and 1,405 registered nurses (2024); 16 hospitals, 147 sub-district health centres

Sources: NSO Nan 2025 [1][2]; RFD 2025 [12]

Mountains, forest and a river head

High ranges in the north, where the Nan River and its tributaries rise, open onto the Nan valley towards the south. Two thirds of the land is classified as forest.

15 districts at a glance, 2024

District	Population	km²	/km²
Mueang Nan	81,528	929	88
Wiang Sa	68,142	1,895	36
Pua	63,717	657	97
Tha Wang Pha	49,847	702	71
Phu Phiang	35,701	392	91
Na Noi	31,681	1,408	23
Chiang Klang	26,514	277	96
Thung Chang	18,893	761	25
Mae Charim	16,422	998	16
Santi Suk	15,471	417	37
Bo Kluea	15,395	848	18
Na Muen	13,854	786	18
Song Khwae	12,392	544	23
Ban Luang	11,180	338	33
Chaloem Phra Kiat	9,770	519	19
Province	**470,507**	**11,472**	**41**

Land use, 2023

Share of 7.17 million rai



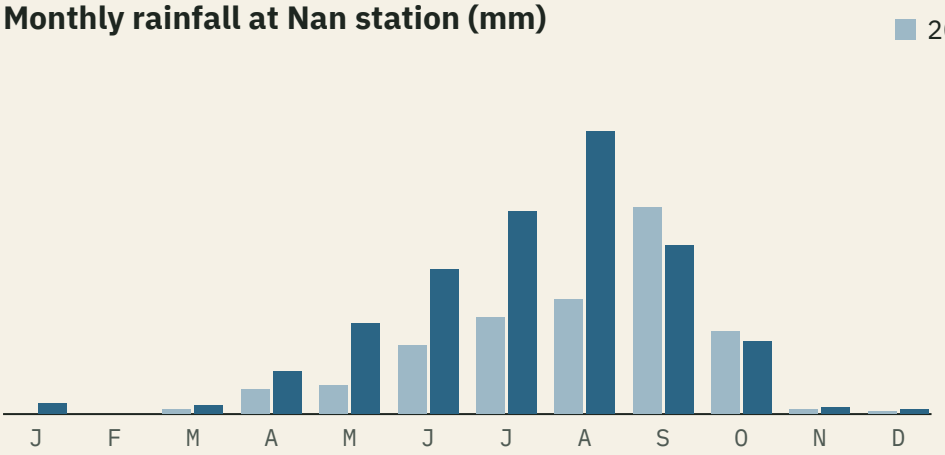
Forest land	**64.6%**
Agricultural land	**20.4%**
Other / unclassified	**15.0%**

Agricultural land (1.47 M rai) by use

Upland field crops	**43.9%**
Fruit & perennial trees	**30.0%**
Paddy	**23.0%**
Other	**2.5%**
Vegetables & flowers	**0.5%**

OAE land-use series. The Royal Forest Department's 2025 satellite estimate, on a 7.58 M rai base, gives 61.0% forest.

Monthly rainfall at Nan station (mm)



471 mm
fell in August 2024 alone, the wettest month of the two years

2024 temperatures: mean 27.6 °C; absolute max 42.0 °C (April); absolute min 13.8 °C

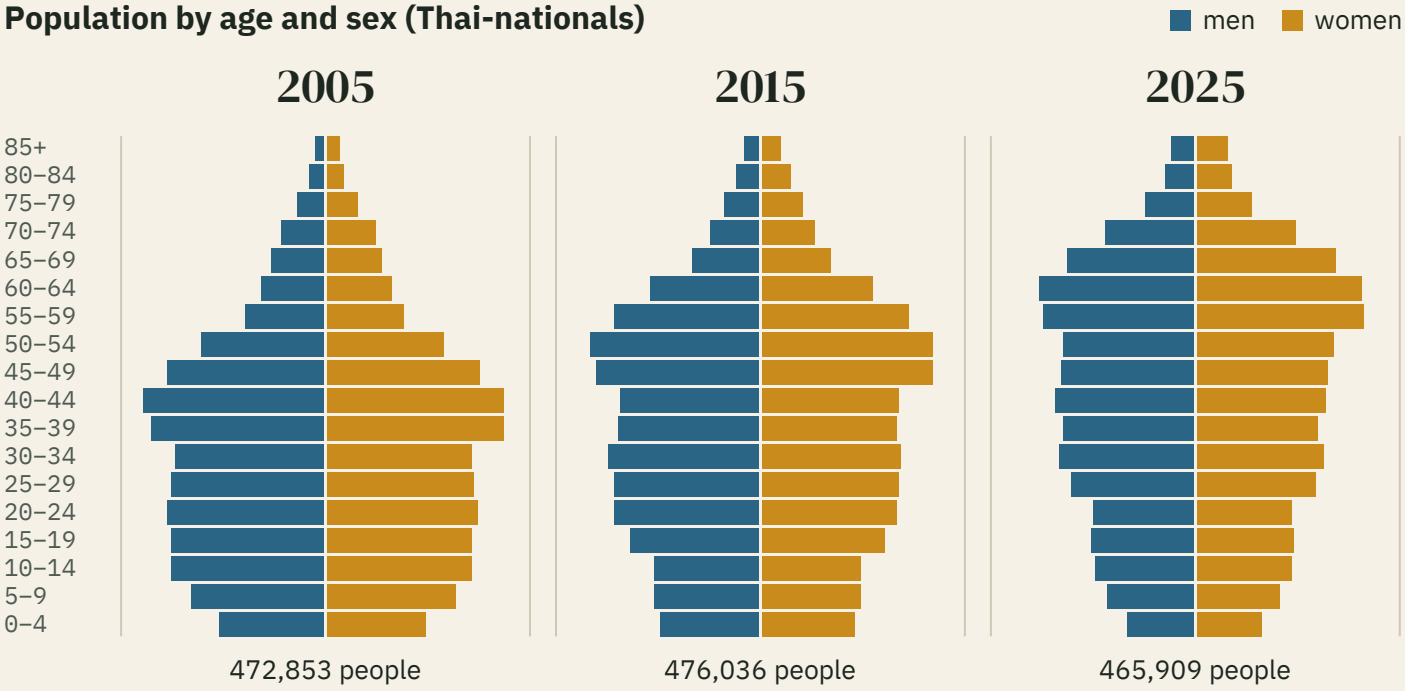
Hot season Mar–Jun · rains Jul–Oct · cool season Dec–Feb

Sources: districts — NSO Nan, Provincial Statistical Report 2025, tables 1.1 & 1.4 (DOPA registration, 31 Dec 2024) [1]; land use — ibid., table 11.1 (OAE) [1]; climate — ibid., tables 20.6 & 20.8 (Nan Meteorological Station) and provincial overview [1]. Density in persons per km².

A shrinking, ageing population

The large generation aged 35–44 in 2005 is now 55–64. Meanwhile births have collapsed: Nan registered 46% fewer infants under one year in 2025 than in 2005.

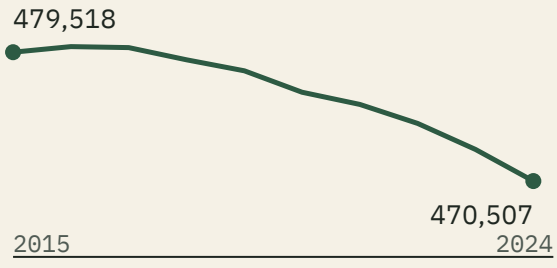
Population by age and sex (Thai-nationals)



Key indicators

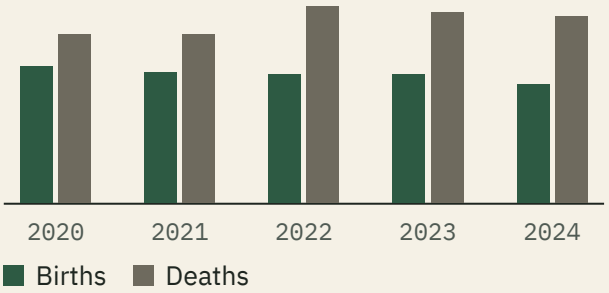
	2005	2015	2025
Share aged 60 and over	11.3%	16.0%	**26.4%**
Share aged 80 and over	1.3%	2.2%	**3.1%**
Share aged under 15	20.1%	15.6%	**13.2%**
Median age (years)	33	38	**44**
People 60+ per 100 aged 15–59	16	23	**44**
Infants under 1 year	4,859	4,197	**2,619**

Registered population, 2015–2024



−9,011 people (−1.9%) in nine years; −0.47% in 2024 alone.

Births and deaths



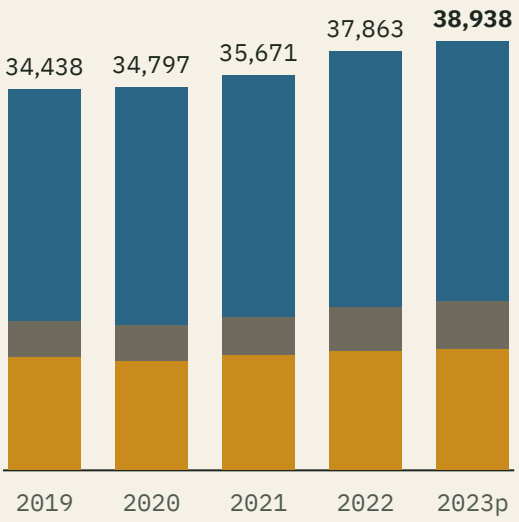
2024: 2,631 births, 4,199 deaths — crude birth rate 5.6‰, death rate 8.9‰.

Sources: Department of Provincial Administration, Nan, December 2005, 2015, 2025 [24] — NSO HSES 2025 [22]; Provincial Statistical Report 2025; Nan Provincial Health Office) [1];

A farm-and-services economy

Agriculture produces more than a quarter of provincial output and employs most workers. Household incomes, below northern and national averages in 2023, rose strongly in 2025. Poverty has fallen sharply over the decade.

Gross provincial product (million-baht, current prices)



89,515 ฿

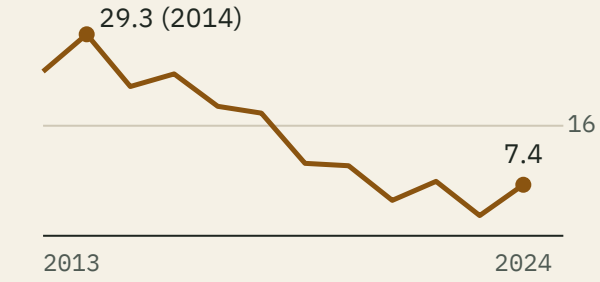
GPP per head, 2023 — 62nd of 77 provinces

Average household income (baht/month)



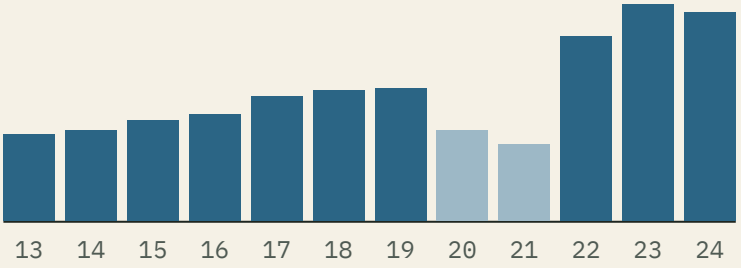
2025 regional and national figures not yet compared. Household finances in detail: p. 6.

Share of people below the poverty line (%)



32,100 poor people in 2024 (consumption-based measure).

Visitors per year (Thai and foreign)



1.51 M visitors in 2024, 1.5% foreign

4,407 M ฿ visitor spending, 2024

70.8% hotel occupancy, 2nd in the North

199

registered factories (2024), 3,250 M baht invested, 1,962 workers; 0.27% of Thailand's factories

Labour force, Q3 2025

0.3% unemployment rate



243,397 employed **1,644** unemployed

123,674 outside the labour force

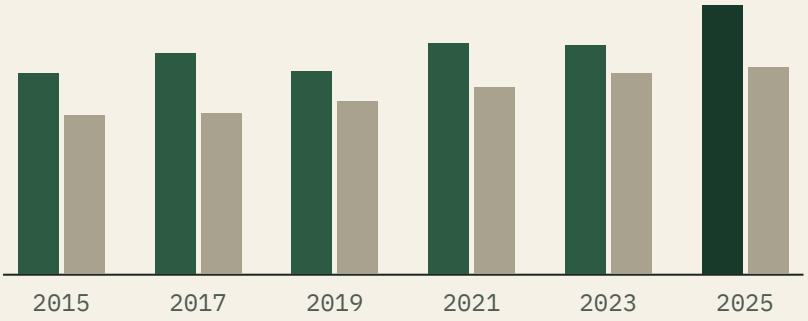
Of 368,715 people aged 15+. Average monthly wage in Q3 2024: 13,805 baht.

Sources: GPP — NESDC via NSO Nan Report 2025 [1]; income 2025 — NSO HSES 2025 [22]; income 2023, poverty, factories, visitors (Ministry of Tourism & Sports, 2013–2024 series) — Nan Pocket Book 2025 [2]; foreign share — Report 2025 [1]; Bars for 2020–21 (pale) = COVID-19 years.

Highest incomes in a decade, debt easing

The 2025 Household Socio-Economic Survey shows falling debt and lower inequality — yet a quarter of income still comes from transfers, not work.

Monthly income and spending per household (฿)



income: 17,886 → **23,904** spending: 14,165 → **18,355**

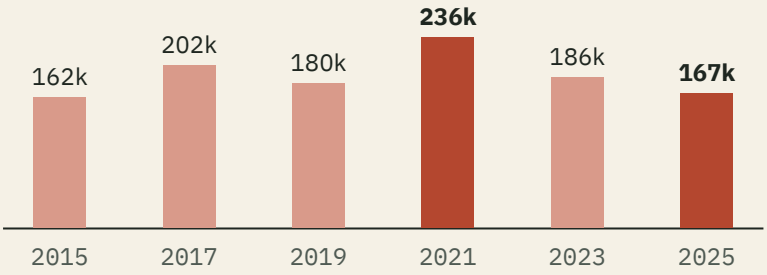
Where 2025 income comes from



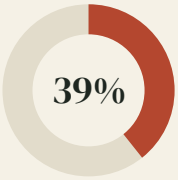
Wages & salaries	**5,471**
Farm profit	**4,561**
Business profit	**2,758**
Transfers & aid	**6,059**
Income in kind	**4,351**
Property & irregular	**704**

Work brings 53.5% of income; farming alone 19%.

Average debt per household (baht)



Averaged over all households. Debt equals 7.0 months of income in 2025, down from 9.2 in 2023.



of households in debt in 2025, down from 46.5% in 2023. Almost all debt is formal; the BAAC is the main lender for 48.7% of indebted households.

What the debt is for (2025)

Consumption	44.5%
Farming	21.5%
House or land	16.9%
Business	15.1%

0.266
Gini coefficient of household income, 2025 (0.305 in 2023)

7,314 ฿
monthly food, drink & tobacco spending — the largest item

5,549 ฿
average monthly surplus of income over spending

Enterprises — 2022 Business & Industrial Census

25,375 active establishments (2021)

Manufacturing	38.9%
Retail trade	25.4%
Other services	9.0%
Food & drink services	8.7%
All other activities	18.0%

99.3%
have 1–15 workers; only 16 have more than 50

95.3%
are owned by individuals

50,371
people working in them

9,552
of 34,927 units listed had closed, paused or moved

Sources: NSO, Household Socio-Economic Survey 2025 — Nan Province (published 2026) [22]; NSO Nan, 2022 Business and Industrial Census, Nan [23].

107,796 farm holdings

The 2023 Agricultural Census portrays small, mostly crop-based farms, widely mechanised, heavily indebted and exposed to rising input costs and drought.

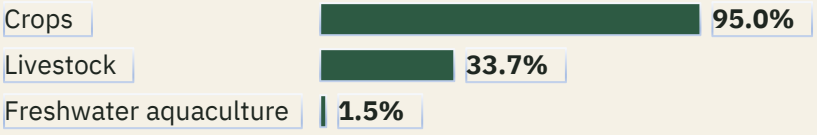
1.76 M rai
farmed by holders (≈282,000 ha)

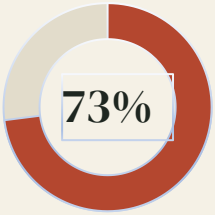
16.3 rai
average holding (≈2.6 ha)

98.2%
of holders on the farmer register

35.4%
use farming apps (to register with government, check weather)

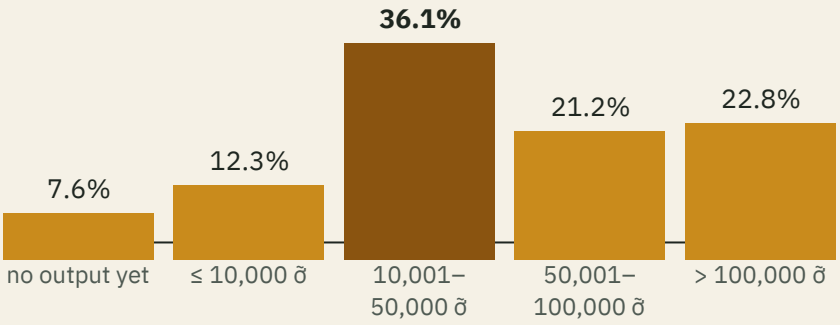
What holders do (several answers possible)

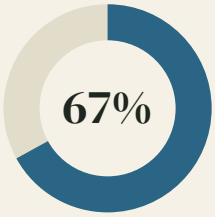




of holders are in debt (63,066). Average debt per farm household: **209,359 ฿**

Annual farm income per holding

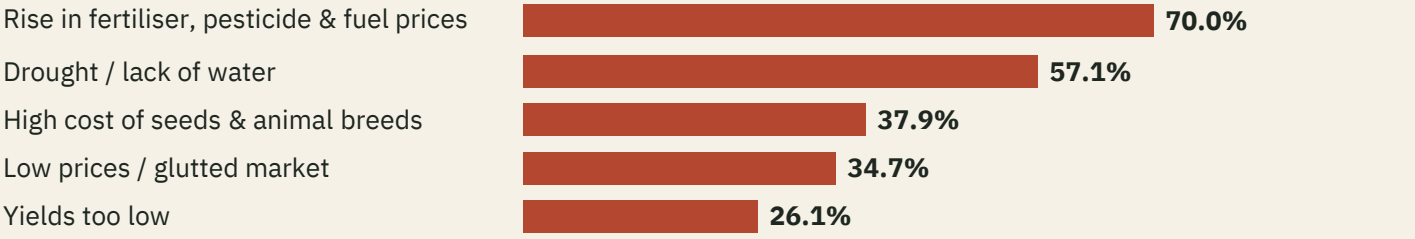




use machinery (54,463). Most common: 4-wheel tractor (54.9% of users), hand sprayer (46.9%), hand weeder (29.3%).

Top five problems reported by farm holders

74.3% reported at least one problem or disaster



What villages ask for, 2024 survey

52.5%
solutions to farming problems

28.1%
household water supply

25.6%
land title documents

19.3%
road repair

18.4%
debt relief

Sources: NSO, Agricultural Census 2023 — Nan results (infographic) [3], reproduced in Nan Pocket Book 2025, p. 12 [2]; village needs — NSO survey of village problems and needs 2024, Pocket Book p. 10 [2]. 1 ha = 6.25 rai.

Maize, rice and rubber cover the land

Three crops account for most of the planted area. Rice of all kinds — lowland, dry-season and upland — covers about as much land as feed maize.

Planted area, crop year 2024 (rai)

field crops · rice · tree & fruit crops

Crop	Area	Output (t)
Feed maize	484,052	329,644
Lowland rice (wet season)	332,059	166,483
Para rubber	329,606	47,675
Upland rice	121,309	46,343
Cassava	78,188	249,241
Longan	60,774	27,595
Dry-season rice	26,876	15,496
Mango	18,618	8,328
Coffee	12,945	7,795
Lychee	12,196	2,415
Rambutan	9,970	3,514
Ginger	3,415	10,311
Tangerine	3,196	3,092
Oil palm	3,048	1,914

480,244 rai
of rice in all forms; glutinous (sticky) rice makes up 57% of wet-season lowland rice

GI
"Som Si Thong" golden orange holds a geographical indication

332,520 rai
of rubber standing in 2025, 274,346 rai tapped (OAE–DOAE agreed figures); 34,203 growers have mapped 355,478 rai of registered plots (July 2026)

+13%
coffee area 2022→2024 (11,484 → 12,945 rai), listed with sericulture and herbs among Nan's high-value products

Livestock, 2024 · 60,053 livestock farmers

1.92 M chickens
77,414 pigs
68,218 ducks
63,689 beef cattle
10,139 buffaloes
3,454 goats & sheep

Inland fisheries, 2024

2,926 t
from 11,241 fish farms (ponds, rice-fish, cages)

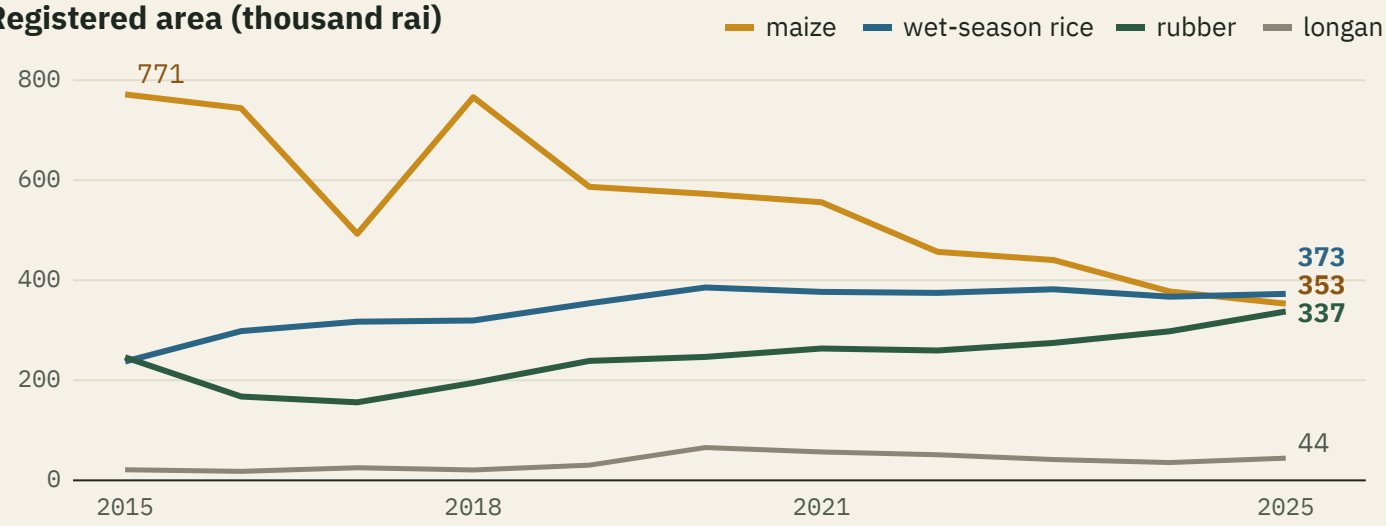
1,498 t
wild freshwater catch

Sources: NSO Nan, Provincial Statistical Report 2025 (Nan Provincial Agricultural Extension, Livestock and Fishery Offices) [1]; coffee 2022 — Report 2023 [4]. Figures come from farmer registration and may not be exhaustive (Nan Provincial Agricultural Office, 2026). High-value products: Pocket Book [2]. Rubber 2025: OAE–DOAE data-unification tables [26]; plot registration GIS summary, 15 Jul 2026 [27].

A decade of crop switching

Farmer-registration records show registered maize land more than halving between 2015 and 2025, while rubber and rice grew. Registration follows support schemes, so shifts partly reflect who registers, not only what is planted.

Registered area (thousand rai)



−54% maize area; growers 34,036 → 20,167

+57% wet-season rice area

+37% rubber area; growers 22,391 → 24,474

×2.1 longan area

Registered maize by district (thousand rai)

District	pale 2015 · dark 2025	change
Wiang Sa		−32%
Na Noi		−41%
Mae Charim		−72%
Tha Wang Pha		**−93%**
Mueang Nan		−66%
Na Muen		−77%
Santi Suk		−79%
Phu Phiang		−86%
Pua		−73%
Chaloem Phra Kiat		−52%
Song Khwae		−77%
Ban Luang		−29%
Bo Kluea · Chiang Klang · Thung Chang	each under 7,000 rai in both years	

Wiang Sa alone held 48% of registered maize in 2025 (170,924 rai).

Farm support at a glance, 2024

63,840 rai
irrigated area; 83 small and medium reservoirs hold 68.7 million m³

137 groups
"large-plot" collective farming schemes, 28 commodities, 9,478 farmers

7,387 farmers
GAP-certified for crops (34,178 rai); 230 farmers organic under PGS (1,471 rai)

1,599 · 517
Smart Farmers · Young Smart Farmers; 15 main and 176 network farmer learning centres

Sources: DOAE farmer-registration data, Nan, 2015–2025 [25]; farm support — Nan Provincial Agriculture and Cooperatives Office, "Agricultural situation and development directions, Nan 2024" [29].

The maize boom and the forest

Feed maize spread quickly up Nan's slopes in the 2000s, driving forest loss. Its area has been shrinking in recent years, while the province experiments with new land-use arrangements.

Seen from space, 1990 → 2019

Satellite classification, km²; panels use different scales



Para rubber grew 6.5-fold between 2007 and 2019; satellite maize area fell 17.6% in 2014–2019. In all, 1,298 km² of forest changed cover over the 30 years (map on the cover) [21].

Official maize-area figures differ by agency (e.g. 2023: 504,728 rai in NSO tables, 564,562 rai by OAE Region 2 [1][20]). The Commerce Office estimated 372,090 rai for 2025/26 [5].

≈1.2 M rai

of forest lost in Nan between 1964 and 2018 [9]

Average forest loss per year (rai)



66,072 ha

of forest lost 2001–2016 (9.1% of the 2000 cover); about 92% became cropland, mainly maize. Annual loss peaked in 2012 (7,523 ha), the year world maize prices peaked [8].

Forest area today

4.63 M rai

forest in 2025 = 61.0% of the province [12]

−10,385 rai

change 2024→2025 (−0.22%) [12]

Land legally designated as forest (6.50 M rai)



2.78 M rai conservation forest

3.71 M rai 17 national reserved forests

Much farmland and many settlements lie inside these legal forest zones; resolving the resulting disputes is a stated aim of the Nan Sandbox [10]. In 2024 GISTDA mapped 2.70 M rai (35.5% of the province) as farmland [13].

The "Nan Sandbox" model

A public–private land-management scheme for the province's forest reserve, co-chaired by the Ministry of Natural Resources and the private sector, operating in all 15 districts (924 villages, 2019 figure) [10].



72% remains watershed forest reserve

18% farmland restored to large trees, with crops allowed under them

10% left to economic crops, still legally forest reserve

139,522

farmers cultivating inside reserved or conservation forest land, as counted in the provincial plan [28]

84,429 rai

allocated to communities under the national land policy (CTC), of 124,034 rai approved by the RFD and 273,663 rai requested [29]

301

registered community forests in 14 districts [28]

Sources: [1] NSO Nan Report 2025; [5] Nan Commerce Office, 10 Oct 2025; [8] Zeng, Gower & Wood 2018; [9] Khemphet & Jongkaewwattana 2021; [10] Thai Post, 14 Feb 2019; [12] RFD 2025; [13] Naewna/GISTDA, 22 Aug 2026; [20] OAE, 1 Sep 2023; [21] Kruasilp et al. 2023; [28] Provincial plan, FY2027 review; [29] Nan Agriculture & Cooperatives Office 2024.

Water tower, smoke and floods

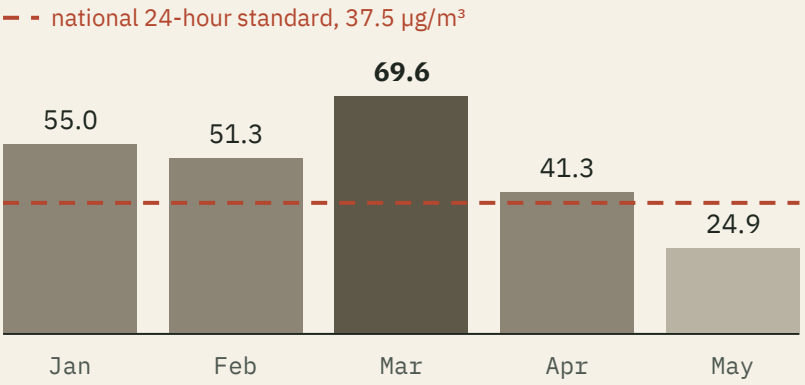
What happens on Nan's slopes is felt downstream: the province heads the main tributary of the Chao Phraya. Locally, dry-season haze and wet-season floods are the recurrent hazards.

≈40%
of the Chao Phraya's runoff comes from the Nan River basin (about 12,000 million m³ a year)

≈57%
of the Chao Phraya's sediment load comes from the same basin

≈34,680 km²
Nan River basin studied, about three times the province's area

PM2.5, monthly average, Jan–May 2025 (µg/m³)



Daily peak: **122.9 µg/m³** on 24 March 2025 — more than three times the national 24-hour standard.

Villages at very high risk, 2024

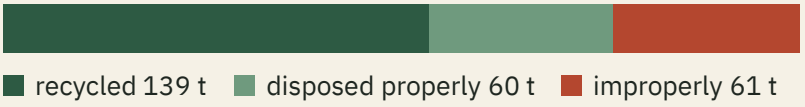
	districts	subdist.	villages
Flood	8	22	**51**
Drought	10	15	**43**
Storm	8	14	**41**

DDPM risk dashboard. Drought is also the second problem cited by farm holders (57.1%, p. 7).

4,847
hotspots a year, avg 2021–25; target −10%

501,738 rai
burned a year, avg 2023–25; target −20%

Solid waste, 2023 — 260 tonnes a day



−29,059 rai

forest change 2015–2025, concentrated in Chaloem Phra Kiat and Thung Chang, the upper Nan watershed

AGRICULTURE & COOPERATIVES PLAN

“Creative, high-value Nan agriculture; farmers happy and sustainable”

Aim: lift farmers out of the middle-income trap. Indicators include agricultural GPP +0.5% and farm-household net income +5% a year.

Threats named in the plan

— Climate change: haze, drought, erratic rain, floods

— Forest and land laws not yet fully implemented

— Steep terrain raising farm costs per rai

Sources: Namsai et al. 2024, Water 16(1):148 [16]; PM2.5 — PM2.5 monitoring system (GISTDA and partners) in Nan Pocket Book 2025 [2]; standard — National Environment Board, ONEP 2022 [19]; risk villages — DDPM dashboard, in Pocket Book [2]; hotspots and burned area — baselines in the provincial plan, FY2027 review [28]; floods — TNN/TrueID 21 & 23 Aug 2026 [14][15], Naewna 22 Aug 2026 [13]; waste — Pollution Control Dept [2]; forest change — RFD data via Naewna/GISTDA [13].

How the province reads its farm sector

The provincial plan's SWOT analysis and the agriculture office's own diagnosis converge on water, land rights, chemical use, ageing farmers and weak processing.

STRENGTHS

— Climate and land suited to high-value products: mango, coffee, longan, orange, tea, herbs;

— Many farmer groups: processing groups, community enterprises, Young Smart Farmers

— Collection centres and processing plants with GMP standards

WEAKNESSES

— Inefficient water management; insecure land documents limit access to production inputs

— Inappropriate chemical use; poor planning along the production chain

— Processing without standards; market information not used, so supply overshoots demand

OPPORTUNITIES

— Growing health consciousness and online, digital sales channels

— The Laos–China railway opens new markets towards China and other countries

— State policies on land allocation (CTC), land titles and royal projects

THREATS

— Climate change bringing plant diseases, animal diseases, pests and disasters

— Epidemics disrupting logistics and markets

— Land and forest laws: farmers without land documents cannot obtain organic or GAP certification

Provincial Development Plan 2023–2027

ISSUE 1 · PEOPLE

Good health, opportunity and a peaceful society

Support people of all ages for good health and a peaceful society.

ISSUE 2 · ECONOMY

A creative, high-value economy

Build infrastructure for farming, water, trade, and tourism

ISSUE 3 · ENVIRONMENT

Environmentally friendly growth

Restore natural resources through public participation.

Selected targets and their baselines

Indicator	Target	Baseline
Life expectancy	**78 years**	77.4 years (2023)
Human achievement index	**≥ 70%**	64% (index 0.6496, 2022)
Population below the poverty line	**down 10%**	11.2% (2020)
Gross provincial product, current prices	**+1.5% a year**	38,938 M ฿ (2023)
Value of GAP and organic produce	**+4% a year each**	+3% (2021)
Sales of Nan identity products	**+4%**	2,190 M ฿ (2024)
Forest area, all types	**+0.16%**	+7,536 rai in 2024
Fire hotspots	**–10% vs 5-yr average**	4,847 a year (2021–25)
Burned area	**–20% vs 3-yr average**	501,738 rai a year (2023–25)

Sources: SWOT — Nan Provincial Development Plan 2023–2027, FY2027 review (agriculture) [28]; Nan Provincial Agriculture and Cooperatives Office 2024 [29]; Nan Pocket Book 2025 [2].

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Notes on the data

Years are Common Era (Thai B.E. − 543). 1 ha = 6.25 rai; 1 km² = 625 rai.

Crop areas come from farmer registration (DOAE), surveys (OAE), satellite mapping (GISTDA, research) or field estimates (Commerce). Each measures something different, so trends are more reliable than exact levels. Interpolated or estimated values found in preparatory files were not used.

Compiled October 2026. Figures are reproduced as published; only unit conversions, shares, medians and changes were computed from the cited tables.